

CLASS 8 · CBSE · GANITA PRAKASH PART 1 · FORMULA BANK

## Chapter 2: Power Play

Every formula and result of the chapter on one sheet

|    |                            |                                                       |
|----|----------------------------|-------------------------------------------------------|
| 1  | Meaning                    | $n^a = n \times n \times \dots \times n$ ( $a$ times) |
| 2  | Product, same base         | $n^a \times n^b = n^{a+b}$                            |
| 3  | Quotient, same base        | $n^a \div n^b = n^{a-b}, n \neq 0$                    |
| 4  | Power of a power           | $(n^a)^b = n^{ab}$                                    |
| 5  | Product, same exponent     | $m^a \times n^a = (mn)^a$                             |
| 6  | Quotient, same exponent    | $m^a \div n^a = \left(\frac{m}{n}\right)^a, n \neq 0$ |
| 7  | Zero exponent              | $n^0 = 1, n \neq 0$                                   |
| 8  | Negative exponent          | $n^{-a} = \frac{1}{n^a}, n \neq 0$                    |
| 9  | Negative base              | Even power positive, odd power negative               |
| 10 | Codes                      | $k$ places with $n$ symbols each give $n^k$ codes     |
| 11 | Scientific notation        | $x \times 10^y$ with $1 \leq x < 10, y$ an integer    |
| 12 | Lakh, crore, arab          | $10^5, 10^7, 10^9$                                    |
| 13 | Million, billion, trillion | $10^6, 10^9, 10^{12}$                                 |